Context – @data26 Capstone Project

Multicenter study on glucose metabolism, risk factors and care pathways

Population: 1,500 adult patients across 24 care centers

1. A flagship exercise at the heart of the @data training program

As part of the @data training program, you will work throughout the core curriculum on the same case study, which will serve as a common thread. The aim is to enable you to progressively put into practice the concepts, methods and tools covered in the different courses, rather than discovering them only through independent exercises.

The dataset presented in this document will serve as the common basis for this flagship project. Through the study of glucose metabolism, diabetes, risk factors and care pathways, you will work with the same patients and the same tables across several training modules, with objectives and tools that will evolve as you progress.

In the following sections, you will discover the study framework, how the data are organized and the main dimensions to be analyzed. You can return to this context document whenever you need to interpret a variable, understand a table or place an analysis in its broader context.

2. General study context

Type 2 diabetes and disturbances in glucose metabolism result from complex interactions between individual characteristics, anthropometric factors, lifestyle habits, medication exposure, family and biological history, as well as medical care. The dataset brings together several of these dimensions for the same patients.

The multicenter study includes 1,500 adults associated with different types of healthcare facilities. The tables combine clinical characteristics, repeated biological measurements, a lifestyle questionnaire, family information, gene expression values and care pathway data.

The dataset is used for educational purposes and is fictional.

2.1 Overall research question

The general question is: which individual, biological, familial, behavioral, medication-related and care pathway factors are associated with the glycemic profile and its evolution over time in the available data?

This deliberately broad question can be broken down into several analyses: determinants of high blood glucose, factors associated with HbA1c, changes in repeated measurements, the relationship between lifestyle and metabolism, the role of corticosteroids, associations with family history, exploration of gene expression profiles, and description of medical care and healthcare utilization.

3. Population and data collection framework

The provided version includes 1,500 adult patients, identified from P001 to P1500. The identifier links the tables at the patient level. Tables describing consultations, treatments, family history or repeated measurements contain several rows per patient.

Patients are associated with 24 care centers identified by codes C01 to C24 and described by their region and facility type. Each patient is linked to a single center in Patients_centres. Information on recruitment procedures and the administrative identity of the centers is not included in the files.

3.1 Periods covered by the data

The files cover different periods: consultations extend from 2023 to June 2026, treatments start as early as December 2022, and most glycemic measurements date from 2024 to June 2026. The 46 patients from center C21 also have historical measurements from 2020 to 2023. Individual characteristics and lifestyle data are not dated in their respective tables.

3.2 Data sources

Data source	Main information	Timing
Clinical record	Age, sex, weight, height, blood group, blood glucose, insulin levels, corticosteroids	Reference time point
Lifestyle questionnaire	Stress, sleep, physical activity, diet, smoking, alcohol	Reference time point
Biological and clinical follow-up	Blood glucose, weight, blood pressure, HbA1c	Repeated measurements
Molecular component	Normalized expression of 19 genes related to diabetes/metabolism	One measurement per patient
Family interview	Relationship and family conditions	History
Prescription / treatment	Medication, start and end dates, dosage	Longitudinal
Consultation record	Date, type, service/physician, reason	Longitudinal
Center reference	Center, region, facility type	Structural

4. Dataset organization

The dataset is organized in relational form. Some tables contain exactly one row per patient, while tables describing events or repeated measurements may contain several rows for the same patient. This distinction is essential for merging operations and for selecting appropriate statistical methods.

Table	Unit of observation	Content
Patients	1 row/patient	Demographic, anthropometric and baseline biological characteristics
Lifestyle	1 row/patient	Stress, sleep, physical activity, diet, smoking, alcohol
Glucose history	Several rows/patient	Measurement dates, blood glucose, weight, blood pressure, HbA1c
Gene expression	1 row/patient	Expression of 19 genes related to diabetes and metabolism
Family history	1 to several rows/patient	Relationship and family conditions; explicit row if no known family history
Treatments	1 to several rows/patient	Medications, treatment periods and dosage
Consultations	Several rows/patient	Dates, consultation types, service and reason
Care centers	1 row/center	Center identity, region and facility type
Patients and centers	1 row/patient	Link between each patient and their main center

5. Description of the main clinical dimensions

5.1 Glucose metabolism and anthropometry

Blood glucose is expressed in mg/dL and is a central variable in the study. Insulin levels, expressed in µIU/mL, provide additional information on the metabolic profile. Weight is expressed in kilograms and height in centimeters. Body mass index (BMI) can be calculated using the formula: weight (kg) / [height (m)]^2.

The history table contains repeated measurements of blood glucose, weight, blood pressure and HbA1c. It can be used to study trajectories provided that the dates, irregularity of follow-up and the fact that the years 2020–2023 concern only center C21 are taken into account. To compare centers, a common observation period should be defined, for example 2024–2026, or another method for handling this difference should be explicitly stated.

5.2 Lifestyle

Lifestyle habits are represented by a stress score from 0 to 10, average sleep duration, the number of minutes of physical activity per week, several dietary indicators, smoking and alcohol consumption. Derived categorical variables also describe stress level, activity level and diet quality.

These variables were designed to be analyzed either independently or as adjustment factors. In particular, they make it possible to explore the difference between an observed association and an association that persists after accounting for age, sex, BMI, treatments or other individual characteristics.

5.3 Family history

The family history table describes, for each patient, the possible presence of conditions among relatives: type 2 diabetes, hypertension, obesity or cardiovascular disease. The relationship to the relative is also recorded. This structure makes it possible to construct indicators such as the presence of a family history of diabetes or the number of reported familial conditions.

5.4 Gene expression

The molecular component contains, for each patient, an expression value for 19 genes: TCF7L2, PPARG, KCNJ11, ABCC8, SLC30A8, CDKAL1, CDKN2A, IGF2BP2, HHEX, FTO, IRS1, GCK, GCKR, MTNR1B, KCNQ1, ADCY5, GLIS3, JAZF1 and HNF1A.

6. Treatments and corticosteroid exposure

The treatment table contains a medication, its category, a start date, an end date when provided, and a dosage. The categories present are antidiabetic drugs, corticosteroids, antihistamines, antibiotics and antidepressants, as well as “No treatment”. A blank end date means that the treatment is still being taken.

7. Care pathways and consultations

The consultations table describes dated healthcare events. It distinguishes, in particular, general practice, preventive care, dietetics, endocrinology and diabetes follow-up consultations. The service or healthcare professional and the reason for the consultation are also documented.

The table makes it possible to describe healthcare utilization, the number and timing of consultations, and to examine their relationship with biological changes. More frequent consultations may be related to health status, the type of follow-up or other unobserved factors; the file does not make it possible to determine with certainty which consultation each treatment is associated with.